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Software Engineering

Sheet 6

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Software Engineering Sheet 6

① Software Design is a Creative Process of Identifying Software Components & their relationships Based on Customer requirements
Software Implementation is realizing this Design By Writing executable Programs
These 2 Activities are Inter-Leaved During Development, meaning Design Continuously Influences Implementation and Vice Versa

② Object Oriented Design involves Creating System Models that define Classes, Objects and their Interactions

UML is used as a Standardized Notation to Model these diagrams

Facilitating understanding & Communication among Stakeholders

③ "Build or Buy" refers to the Decision to either Develop Software From Scratch or Purchase off-the-shelf (CoTs) Systems

- Buying is Cheaper & Faster, especially when existing Packages can be Adapted ex → Buying Medical record systems already used in Hospitals, rather than Custom - Building one

- When Buying, the Design Process Focuses on Configuring the System to Meet Specific Requirements rather than Creating it From the ground-up

④ Object-oriented Design Process Stages :

- ① Define the Context of System → understanding the Environment where the system operates
- ② Define the System Architecture → Planning How Components Interact
- ③ Identify the System Objects → Finding Key Classes and Objects
- ④ Develop Design Models → Creating UML Diagrams for Structure & Behaviour
- ⑤ Specify Object Interfaces → Defining How Objects Communicate with each other

Example → Weather Station System

⑤ understanding the Relationships Between the System & its Environment

Helps Decide What Functionality is Included in the System and

How to Structure Communication, it helps in Setting System Boundaries, Clarifying Which Features Belong inside System Versus external Systems

⑥ Identifying Object Classes is crucial as these represent the Primary System Entities & their Behaviours
its Approaches :

- ① Grammatical Approach
- ② Tangible Approach
- ③ Behavioural Approach
- ④ Scenario-Based Analysis

- ⑦ Structure Models → Describe the Static Structure of the System
Showing Object Classes & their Relationships

Dynamic Models → Describe How Objects Interact Over Time
Modeling the System's Behaviour & Message Exchanges

⑧ ~~Importance~~ Importance & Benefits of Design Patterns :

- Patterns Capture Best Practices & Reusable Experience for Designing Software
- They Help Solve Design Problems By Applying Proven Solutions

Improving Design Quality & Developer Communication

⑨ Observer Design Patterns :

- Separates the Display of the state of an Object from the Object itself and allows Alternative Displays to be Provided

When the Object State Changes, all Displays are Automatically Notified & updated to reflect the change

its Components :

- Name
- Description
- Problem Description
- Solution Description
- Consequences

⑩ Open-Source Development :

Benefits → Promotes Reuse, Collaboration, & Shared Improvements

Challenges → Managing Versions, Contributions & Consistency

- Licenses Affects How Software Can Be used, Shared, & Modified

MCQs

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- ① (B) To Organize Software Components & Their Relationships
- ② (C) Writing the Final Code
- ③ (A) The Decision of Use off-the-shelf Software or Build a custom System
- ④ (C) to Show the Sequences the Object Interactions over Time
- ⑤ (A) they Show How Objects Respond to Service requests & Transitions Between States
- ⑥ to Display the State of an Object
- ⑦ Licensing Complexity
- ⑧ (B) Tracking & Managing Software Versions & System Integrations
- ⑨ (A) Developing Software on One Machine & running it on another
- ⑩ (B) Software whose ~~code~~ Source Code is Publically Available For Modification & redistribution